Fourier Series and the Discrete Fourier Transform

Computational Physics II – Unit 4: Fourier Transform I

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1. Motivation: why Fourier in physics?

- Many physical systems are **linear**: waves, RLC circuits, oscillators, heat diffusion, the free Schrödinger equation.
- In a linear system **superposition** holds: if we know the response to $\cos(\omega t)$ and $\sin(\omega t)$, we know the response to any combination of them.
- The exponentials $\mathrm{e}^{\mathrm{i}\omega t}$ are **eigenfunctions** of the derivative operator:

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathrm{e}^{\mathrm{i}\omega t} = \mathrm{i}\omega\,\mathrm{e}^{\mathrm{i}\omega t}.$$

  So a linear differential equation becomes an *algebraic* equation for each frequency.
- **Central idea:** decompose an arbitrary periodic function into a sum of harmonic oscillations, each with its own amplitude and phase.

Goal of this presentation

Understand where every formula comes from (not just use it), justify every simplification, and take everything to a numerical code with a physical application.

2. Orthogonality of sines and cosines

We work with functions of period 2π on $[-\pi, \pi]$ (we generalize later).

Result (for integers $m, n \geq 1$):

$$\int_{-\pi}^{\pi} \cos mx \cos nx \, \mathrm{d}x = \pi \, \delta_{mn}, \quad \int_{-\pi}^{\pi} \sin mx \sin nx \, \mathrm{d}x = \pi \, \delta_{mn}, \quad \int_{-\pi}^{\pi} \sin mx \cos nx \, \mathrm{d}x = 0.$$

Proof. We use the product-to-sum identity:

$$\cos mx \cos nx = \tfrac{1}{2}\big[\cos(m-n)x + \cos(m+n)x\big].$$

For an integer $k \neq 0$: $\displaystyle\int_{-\pi}^{\pi} \cos kx \, \mathrm{d}x = \frac{2 \sin k\pi}{k} = 0$; for $k = 0$ it equals 2π.

- If $m \neq n$: both $m - n \neq 0$ and $m + n \neq 0$, so both integrals vanish $\Rightarrow$ result 0.
- If $m = n$: $\tfrac{1}{2}[2\pi + 0] = \pi$.

For $\sin \cdot \cos$ the integrand is **odd** (odd times even), and the integral of an odd function over a symmetric interval is zero.

Interpretation

$\{1, \cos nx, \sin nx\}$ is an orthogonal basis of the space of periodic functions, with inner product $\langle f, g \rangle = \int_{-\pi}^{\pi} fg \, \mathrm{d}x$.

3. Computing the Fourier coefficients

Assume f can be written as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos nx + b_n \sin nx \right].$$

To isolate a_m: multiply by $\cos mx$ and integrate over $[-\pi, \pi]$:

$$\int_{-\pi}^{\pi} f \cos mx \, \mathrm{d}x = \frac{a_0}{2} \underbrace{\int \cos mx \, \mathrm{d}x}_{0 \ (m \geq 1)} + \sum_n \Big[a_n \underbrace{\int \cos nx \cos mx \, \mathrm{d}x}_{\pi \delta_{nm}} + b_n \underbrace{\int \sin nx \cos mx \, \mathrm{d}x}_{0} \Big] = \pi a_m.$$

$$\boxed{a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f \cos nx \, \mathrm{d}x, \qquad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f \sin nx \, \mathrm{d}x}$$

For $m = 0$:
$\int f \, \mathrm{d}x = \frac{a_0}{2} \, 2\pi \Rightarrow a_0 = \frac{1}{\pi} \int f \, \mathrm{d}x.$

Why $a_0/2$ and not a_0?

For $m = 0$ the integral of $\cos^2$ is 2π, not π. Writing the constant term as $a_0/2$ makes *the same formula* $a_n = \frac{1}{\pi} \int f \cos nx$ valid for every $n \geq 0$. Also, $a_0/2$ is the **mean value** of f.

Assumption made

Integrating the series term by term: valid if the series converges uniformly (or, in general, in the L^2 sense).

4. Symmetries: when coefficients vanish

Basic rule: $\displaystyle\int_{-\pi}^{\pi} g(x)\,\mathrm{d}x = 0$ if g is odd, and $= 2\int_0^{\pi} g(x)\,\mathrm{d}x$ if g is even.

If f is…	$f\cos nx$ is…	$f\sin nx$ is…
even	even $\Rightarrow a_n = \frac{2}{\pi}\int_0^{\pi} f\cos nx$	odd $\Rightarrow b_n = 0$
odd	odd $\Rightarrow a_n = 0$	even $\Rightarrow b_n = \frac{2}{\pi}\int_0^{\pi} f\sin nx$

Why this simplification is legitimate

It is not an approximation, it is an exact equality. It follows from (even)×(even)=even, (odd)×(odd)=even, (even)×(odd)=odd, plus the change of variable $x \to -x$ on the negative half of the interval: $\int_{-\pi}^{0} g(x)\,\mathrm{d}x = \int_0^{\pi} g(-x)\,\mathrm{d}x$.

Practical benefits:

- The integration interval is halved (less work).
- You know *in advance* which coefficients must be zero: an excellent **check** of numerical results.
- An extra symmetry ($f(x+\pi) = -f(x)$, "half-wave") makes **only odd harmonics survive**.

5. Example 1: the square wave (step by step)

$$f(x) = \begin{cases} -1, & -\pi < x < 0 \\ +1, & 0 < x < \pi \end{cases} \qquad f(x + 2\pi) = f(x).$$

1. f is **odd** $\Rightarrow a_n = 0$ for all $n \geq 0$ (including a_0: zero mean value).
2. Coefficients b_n (by symmetry we integrate only on $[0, \pi]$, where $f = 1$):

$$b_n = \frac{2}{\pi} \int_0^{\pi} \sin nx \, \mathrm{d}x = \frac{2}{\pi} \Big[-\frac{\cos nx}{n} \Big]_0^{\pi} = \frac{2\,(1 - \cos n\pi)}{n\pi} = \frac{2\,[1 - (-1)^n]}{n\pi}.$$

3. Since $(-1)^n = -1$ for odd n and $+1$ for even n:

$$b_n = \begin{cases} \dfrac{4}{n\pi}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases} \quad \Longrightarrow \quad f(x) = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\sin\big((2k-1)x\big)}{2k-1}.$$

Check: Leibniz series

At $x = \pi/2$ the series gives $1 = \frac{4}{\pi}\big(1 - \frac{1}{3} + \frac{1}{5} - \cdots\big)$, i.e. $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ (a known result). Even harmonics are missing because $f(x + \pi) = -f(x)$.

6. Example 2: the triangle wave $f(x) = |x|$

On $[-\pi, \pi]$, $f(x) = |x|$, extended with period 2π. It is **even** $\Rightarrow b_n = 0$.

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x\,\mathrm{d}x = \pi, \qquad a_n = \frac{2}{\pi}\int_0^{\pi} x\cos nx\,\mathrm{d}x = \frac{2}{\pi}\Big[\frac{x\sin nx}{n} + \frac{\cos nx}{n^2}\Big]_0^{\pi} = \frac{2\left[(-1)^n - 1\right]}{\pi n^2}.$$

(Integration by parts with $u = x$, $\mathrm{d}v = \cos nx\,\mathrm{d}x$.) Hence

$$a_n = \begin{cases} -\dfrac{4}{\pi n^2}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases} \qquad \Longrightarrow \qquad |x| = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=1}^{\infty} \frac{\cos\left((2k-1)x\right)}{(2k-1)^2}.$$

Two independent checks

- $x = 0$: $0 = \frac{\pi}{2} - \frac{4}{\pi}\sum \frac{1}{(2k-1)^2} = \frac{\pi}{2} - \frac{4}{\pi}\cdot\frac{\pi^2}{8} = 0$ ✓
- Parseval: $\frac{\pi^2}{4} + \frac{1}{2}\sum a_n^2 = \frac{\pi^2}{4} + \frac{\pi^2}{12} = \frac{\pi^2}{3} = \frac{1}{2\pi}\int x^2$ ✓

Key difference with the square wave: here f is *continuous*; only its derivative jumps. The coefficients decay as $1/n^2$ (instead of $1/n$) and there is **no Gibbs overshoot**.

Dirichlet conditions (sufficient): within one period f is bounded, has finitely many maxima/minima and finitely many finite discontinuities. Then the series converges to $f(x)$ at continuity points and to the **average of the one-sided limits** at a jump (here 0 at $x = 0$).

Gibbs: the overshoot does not go away

Near a jump of size Δ, partial sums overshoot the limit by $\approx 9\%\,\Delta$ *no matter how many terms are used*; it only becomes narrower.

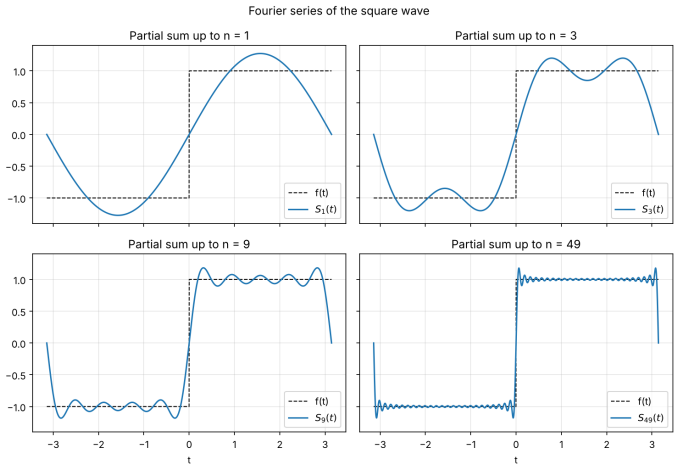
Derivation (partial sum with K odd harmonics).

$$S'_K(x) = \frac{4}{\pi}\sum_{k=1}^{K}\cos(2k-1)x = \frac{2}{\pi}\,\frac{\sin 2Kx}{\sin x} \quad \Rightarrow \quad \text{first maximum at } x_K = \frac{\pi}{2K}.$$

$$S_K(x_K) = \frac{2}{\pi}\int_0^{x_K}\frac{\sin 2Ku}{\sin u}\,\mathrm{d}u \xrightarrow{u=v/2K,\ \sin u\approx u} \frac{2}{\pi}\int_0^{\pi}\frac{\sin v}{v}\,\mathrm{d}v = \frac{2}{\pi}\,\mathrm{Si}(\pi) \approx 1.1790.$$

Why $\sin u \approx u$ is justified: in the integration range $u \le \pi/2K$, which is tiny for large K, so the relative error is $O(u^2) \to 0$. That is why the limit is exact as $K \to \infty$ and independent of K.

Figure: partial sums of the square wave



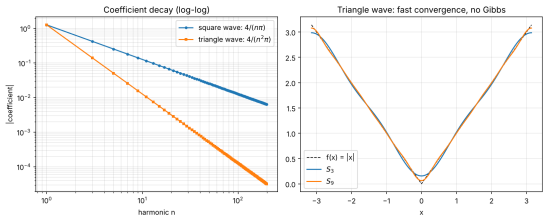
With $N = 49$ the approximation is good away from the jump, but the overshoot near $x = 0$ is still ≈ 0.179.

8. Smoothness and coefficient decay

Claim. If f and its first $p-1$ derivatives are continuous and $f^{(p)}$ is piecewise continuous, then $c_n = O(1/n^{p+1})$. **Derivation** (integration by parts, T-periodic f, $\omega_0 = 2\pi/T$):

$$c_n = \frac{1}{T}\int_0^T f\,\mathrm{e}^{-\mathrm{i}n\omega_0 t}\,\mathrm{d}t = \underbrace{\frac{1}{T}\Big[\frac{f\,\mathrm{e}^{-\mathrm{i}n\omega_0 t}}{-\mathrm{i}n\omega_0}\Big]_0^T}_{=0 \text{ if } f \text{ continuous \& periodic}} + \frac{1}{\mathrm{i}n\omega_0}\underbrace{\frac{1}{T}\int_0^T f'\,\mathrm{e}^{-\mathrm{i}n\omega_0 t}\,\mathrm{d}t}_{c_n(f')}.$$

- Each continuous derivative contributes one factor $1/n$.
- Square wave: $p = 0 \Rightarrow 1/n$.
- Triangle wave: $p = 1 \Rightarrow 1/n^2$.
- Analytic functions: faster than any power (exponential decay).



Practical meaning

Smoother $\Rightarrow$ fewer harmonics needed. With 5 harmonics ($N = 9$) the triangle wave error is 0.064 everywhere; for the square wave it is 0.182 even away from the jumps.

9. Arbitrary period T

If $f(t+T) = f(t)$, define $\omega_0 = 2\pi/T$ and change variable $x = \omega_0 t$ (so $x \in [-\pi, \pi]$ when $t \in [-T/2, T/2]$).

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right]$$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos(n\omega_0 t)\, \mathrm{d}t, \qquad b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin(n\omega_0 t)\, \mathrm{d}t.$$

Where does the factor $2/T$ come from?

With $x = \omega_0 t$, $\mathrm{d}x = \omega_0\, \mathrm{d}t$, so $\frac{1}{\pi}\, \mathrm{d}x = \frac{\omega_0}{\pi}\, \mathrm{d}t = \frac{2}{T}\, \mathrm{d}t$.

Physical meaning: harmonic n oscillates at frequency $f_n = n/T$ (fundamental $f_1 = 1/T$). A long period means harmonics very close in frequency: $\Delta f = 1/T$. This relation will return as the **frequency resolution** of the DFT.

10. Complex form of the series

Using Euler: $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$, $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$. Then

$$a_n \cos n\omega_0 t + b_n \sin n\omega_0 t = \underbrace{\frac{a_n - ib_n}{2}}_{c_n} e^{in\omega_0 t} + \underbrace{\frac{a_n + ib_n}{2}}_{c_{-n}} e^{-in\omega_0 t}.$$

$$\boxed{f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega_0 t}, \qquad c_n = \frac{1}{T}\int_0^T f(t)\, e^{-in\omega_0 t}\, dt}$$

- **Complex orthogonality**: $\frac{1}{T}\int_0^T e^{i(m-n)\omega_0 t}\, dt = \delta_{mn}$. c_n is isolated as before, multiplying by $e^{-in\omega_0 t}$.
- Relations: $c_0 = a_0/2$, $a_n = 2\,\mathrm{Re}\, c_n$, $b_n = -2\,\mathrm{Im}\, c_n$ (for $n \geq 1$).
- **If f is real**, $c_{-n} = c_n^*$: the negative spectrum carries no new information.

Why use the complex form?

It is more compact (one formula), amplitude and phase sit in a single complex number $c_n = |c_n| e^{i\varphi_n}$, and it generalizes directly to the Fourier transform and the DFT.

11. Parseval's theorem

Compute the "mean power" of a real f: $\frac{1}{T}\int_0^T |f|^2\,\mathrm{d}t$. Substituting the complex series for f and for f^*:

$$\frac{1}{T}\int_0^T f f^*\,\mathrm{d}t = \sum_{m,n} c_m c_n^* \underbrace{\frac{1}{T}\int_0^T \mathrm{e}^{\mathrm{i}(m-n)\omega_0 t}\,\mathrm{d}t}_{\delta_{mn}} = \sum_{n=-\infty}^{\infty} |c_n|^2 = \frac{a_0^2}{4} + \frac{1}{2}\sum_{n=1}^{\infty}\left(a_n^2 + b_n^2\right).$$

Physical interpretation

The (average) energy of the signal is the sum of the energies of each harmonic: there are no "cross terms" because harmonics are orthogonal. The factor $\frac{1}{2}$ is the average of $\cos^2$ or $\sin^2$ over a period.

Check with the square wave ($|f|^2 = 1$, so the left side equals 1):

$$\frac{1}{2}\sum_{n\ \text{odd}} \frac{16}{n^2\pi^2} = \frac{8}{\pi^2}\sum_{n\ \text{odd}} \frac{1}{n^2} = \frac{8}{\pi^2}\cdot\frac{\pi^2}{8} = 1\ \checkmark$$

With $N = 9$ harmonics we already recover 96% of the power; with $N = 999$, 99.96% (result of the code).

12. Application: damped driven oscillator

$$m\ddot{x} + \gamma\dot{x} + kx = F(t), \qquad F(t+T) = F(t), \quad \omega_0^2 = k/m, \quad \beta = \gamma/m, \quad \omega = \tfrac{2\pi}{T}.$$

Write $F(t) = \sum_n F_n \, \mathrm{e}^{\mathrm{i}n\omega t}$ and look for $x(t) = \sum_n x_n \, \mathrm{e}^{\mathrm{i}n\omega t}$ (steady state). Since $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}^{\mathrm{i}n\omega t} = \mathrm{i}n\omega \, \mathrm{e}^{\mathrm{i}n\omega t}$:

$$m\left(\omega_0^2 - n^2\omega^2 + \mathrm{i}\beta n\omega\right) x_n = F_n \quad \Longrightarrow \quad x_n = \frac{F_n/m}{\omega_0^2 - n^2\omega^2 + \mathrm{i}\beta n\omega}.$$

- **Justification**: the equation is linear and the exponentials are linearly independent (orthogonal), so the coefficient of each $\mathrm{e}^{\mathrm{i}n\omega t}$ is matched separately.
- Each harmonic is amplified by the **response function** $\chi(n\omega) = 1/(\omega_0^2 - n^2\omega^2 + \mathrm{i}\beta n\omega)$: resonance when $n\omega \approx \omega_0$.
- For a square-wave force only odd harmonics appear ($F_n \propto 1/n$). If $3\omega \approx \omega_0$, the **third harmonic** excites the resonance even though the fundamental frequency is far away.

13. Application: heat equation on a rod

Rod of length L with ends held at 0: $\partial_t u = \alpha\, \partial_x^2 u$, $u(0,t) = u(L,t) = 0$, $u(x,0) = f(x)$. **Separation of variables** $u = X(x)T(t)$: $X'' = -\lambda X$ with $X(0) = X(L) = 0$ forces

$$X_n(x) = \sin \frac{n\pi x}{L}, \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2, \qquad T_n(t) = \mathrm{e}^{-\alpha \lambda_n t}.$$

Superposition and the initial condition give a **Fourier sine series** (period $2L$, odd extension of f):

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}\, \mathrm{e}^{-\alpha (n\pi/L)^2 t}, \qquad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L}\, \mathrm{d}x.$$

Why a sine series?

The boundary conditions $u = 0$ at both ends select the sines (they vanish at 0 and L); the cosines are discarded. Orthogonality of the sines on $[0, L]$ gives b_n.

Physics: mode n decays with rate $\propto n^2$. High harmonics (sharp features, like Gibbs ripples) vanish first: **diffusion smooths**.

14. From the series to the DFT: discretize the integral

In practice we do not know $f(t)$ continuously, only **samples** (measurements or numerical results):

$$x_m = f(t_m), \qquad t_m = m\,\Delta t = \frac{m\,T}{N}, \qquad m = 0, 1, \ldots, N-1, \qquad f_s = \frac{1}{\Delta t} = \frac{N}{T}.$$

Approximate $c_k = \frac{1}{T}\int_0^T f(t)\,\mathrm{e}^{-\mathrm{i}k\omega_0 t}\,\mathrm{d}t$ with the rectangle rule:

$$c_k \approx \frac{1}{T}\sum_{m=0}^{N-1} x_m\,\mathrm{e}^{-\mathrm{i}k\frac{2\pi}{T}\frac{mT}{N}}\,\Delta t = \frac{1}{N}\sum_{m=0}^{N-1} x_m\,\mathrm{e}^{-2\pi\mathrm{i}\,km/N} \equiv \frac{X_k}{N}.$$

Why is the simplest rule good enough?

The integrand is **periodic**: its values at $t = 0$ and $t = T$ coincide. For periodic, smooth integrands the rectangle rule equals the trapezoid rule, and the endpoint error terms of the Euler–Maclaurin formula cancel between the two ends. The error decays faster than any power of $1/N$ (spectral convergence).

This finite sum is, by definition, the **DFT**: $X_k = \sum_{m=0}^{N-1} x_m\,\mathrm{e}^{-2\pi\mathrm{i}km/N}$.

15. Definition and discrete orthogonality

Let $W_N = \mathrm{e}^{-2\pi \mathrm{i}/N}$ (primitive N-th root of unity: $W_N^N = 1$).

$$\boxed{X_k = \sum_{m=0}^{N-1} x_m \, W_N^{km}, \qquad k = 0, 1, \ldots, N-1}$$

Lemma (discrete orthogonality). For integer p:

$$\sum_{m=0}^{N-1} \mathrm{e}^{2\pi \mathrm{i}\, pm/N} = \begin{cases} N, & p \equiv 0 \pmod N \\ 0, & \text{otherwise.} \end{cases}$$

Proof. If $p \equiv 0$, each term is 1 and there are N of them. Otherwise, with $r = \mathrm{e}^{2\pi \mathrm{i} p/N} \neq 1$ it is a geometric series:

$$\sum_{m=0}^{N-1} r^m = \frac{1 - r^N}{1 - r} = \frac{1 - \mathrm{e}^{2\pi \mathrm{i} p}}{1 - r} = 0, \quad \text{because } \mathrm{e}^{2\pi \mathrm{i} p} = 1 \text{ for integer } p.$$

Meaning

The N vectors $\{\mathrm{e}^{2\pi \mathrm{i} km/N}\}_{k=0}^{N-1}$ are orthogonal in $\mathbb{C}^N$: the DFT is a **change of basis** (a unitary rotation up to the factor $\sqrt{N}$), just like the Fourier series is in function space.

$$\boxed{x_m = \frac{1}{N}\sum_{k=0}^{N-1} X_k\, \mathrm{e}^{+2\pi \mathrm{i}\, km/N}}$$

Proof. Substitute the definition of X_k into the right-hand side (dummy index n in the inner sum):

$$\frac{1}{N}\sum_{k=0}^{N-1}\Big[\sum_{n=0}^{N-1} x_n\, \mathrm{e}^{-2\pi \mathrm{i} kn/N}\Big]\mathrm{e}^{2\pi \mathrm{i} km/N} = \sum_{n=0}^{N-1} x_n \underbrace{\frac{1}{N}\sum_{k=0}^{N-1} \mathrm{e}^{2\pi \mathrm{i} k(m-n)/N}}_{\delta_{mn}\ \text{(lemma with } p=m-n)} = x_m.$$

Since $0 \le m, n \le N-1$, $m - n \equiv 0 \pmod N$ only when $m = n$.

Normalization conventions

There are different choices for the $1/N$ factor: (i) in the inverse (NumPy, SciPy); (ii) $1/\sqrt{N}$ in both (unitary transform); (iii) in the forward transform. All are equivalent as long as the *same* one is used in both directions. Here: NumPy's, $X_k = \sum x_m \mathrm{e}^{-2\pi \mathrm{i} km/N}$.

Matrix form: $\mathrm{X} = F\,\mathrm{x}$ with $F_{km} = W_N^{km}$ and $F^{-1} = \frac{1}{N}F^{\dagger}$.

17. A worked example: $N = 4$ by hand

Take $\mathrm{x} = (1,\ 2,\ 0,\ -1)$. Here $W_4 = \mathrm{e}^{-\mathrm{i}\pi/2} = -\mathrm{i}$, so $W_4^0 = 1,\ W_4^1 = -\mathrm{i},\ W_4^2 = -1,\ W_4^3 = \mathrm{i}$ and

$$F = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -\mathrm{i} & -1 & \mathrm{i} \\ 1 & -1 & 1 & -1 \\ 1 & \mathrm{i} & -1 & -\mathrm{i} \end{pmatrix}, \qquad \mathrm{X} = F\mathrm{x} = \begin{pmatrix} 1+2+0-1 \\ 1-2\mathrm{i}+0-\mathrm{i} \\ 1-2+0+1 \\ 1+2\mathrm{i}+0+\mathrm{i} \end{pmatrix} = \begin{pmatrix} 2 \\ 1-3\mathrm{i} \\ 0 \\ 1+3\mathrm{i} \end{pmatrix}.$$

Checks

- **Real signal:** $X_3 = X_1^*$ ✓ (and X_0, X_2 real).
- **Mean:** $X_0/N = 0.5 = \bar{x}$ ✓
- **Discrete Parseval:** $\sum |x_m|^2 = 6$ and $\frac{1}{N}\sum |X_k|^2 = \frac{4+10+0+10}{4} = 6$ ✓

Inverse: e.g.
$x_1 = \frac{1}{4}(2 + (1-3\mathrm{i})\mathrm{i} + 0 + (1+3\mathrm{i})(-\mathrm{i})) = \frac{1}{4}(2+3+\mathrm{i}+3-\mathrm{i}) = 2$ ✓

The code (Part 8) reproduces this and gives zero difference with `np.fft.fft`.

18. Frequencies, Nyquist and aliasing

Since $k\omega_0 = 2\pi k/T$ and $T = N/f_s$, index k corresponds to the frequency

$$f_k = \frac{k}{T} = k\,\frac{f_s}{N}, \qquad \Delta f = \frac{f_s}{N} = \frac{1}{T} \quad \text{(resolution)}.$$

Periodicity in k: $\mathrm{e}^{-2\pi \mathrm{i}(k+N)m/N} = \mathrm{e}^{-2\pi \mathrm{i}km/N}\,\mathrm{e}^{-2\pi \mathrm{i}m} = \mathrm{e}^{-2\pi \mathrm{i}km/N} \Rightarrow X_{k+N} = X_k$. Only N independent values. **Real signal:** $X_{N-k} = \sum x_m \mathrm{e}^{-2\pi \mathrm{i}(N-k)m/N} = \sum x_m \mathrm{e}^{+2\pi \mathrm{i}km/N} = X_k^*$.

Nyquist theorem

With sampling frequency f_s only frequencies $|f| < f_s/2$ (index $k < N/2$) can be represented. A component at $f > f_s/2$ appears *disguised* (alias) at $f - nf_s$.

Example: with $f_s = 100$ Hz, a 93 Hz signal looks like $|93 - 100| = 7$ Hz.

Why: $\cos(2\pi \cdot 93\, t_m) = \cos(2\pi \cdot 7\, t_m)$ at the instants $t_m = m/100$.

- **Rule of thumb**: choose $f_s > 2f_{\max}$ (ideally 5–10 $f_{\max}$) and filter before sampling.
- In NumPy: `np.fft.rfft` (only $k = 0..N/2$, for real signals) and `np.fft.rfftfreq(N, d=1/fs)`.

19. Physical amplitude and phase from the DFT

Let $x_m = A\cos(2\pi f_0 t_m + \varphi)$ with $f_0 = k_0 f_s/N$ (*exact multiple* of Δf, see spectral leakage). Then

$$x_m = \frac{A}{2}\left[\mathrm{e}^{\mathrm{i}\varphi}\mathrm{e}^{2\pi \mathrm{i} k_0 m/N} + \mathrm{e}^{-\mathrm{i}\varphi}\mathrm{e}^{-2\pi \mathrm{i} k_0 m/N}\right].$$

Applying the DFT and the orthogonality lemma (each exponential only "activates" one k):

$$X_{k_0} = \frac{NA}{2}\,\mathrm{e}^{\mathrm{i}\varphi}, \qquad X_{N-k_0} = \frac{NA}{2}\,\mathrm{e}^{-\mathrm{i}\varphi}, \qquad \text{all other } X_k = 0.$$

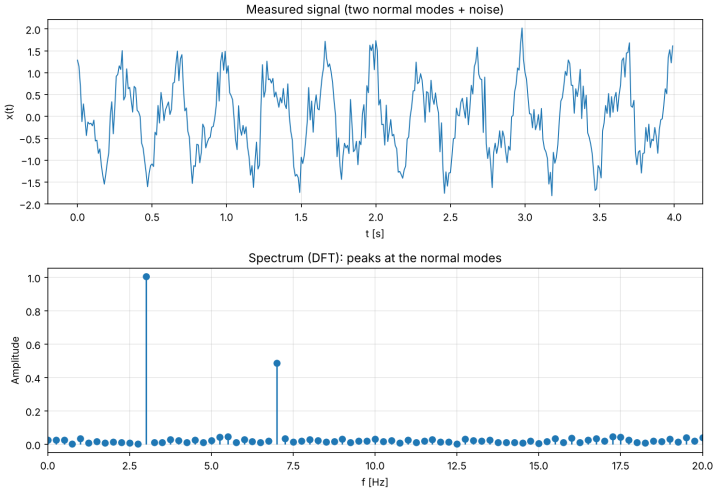
Formulas to read a spectrum

$$A = \frac{2}{N}\,|X_{k_0}|, \qquad \varphi = \arg X_{k_0} \qquad (k_0 \neq 0,\ N/2)$$

The factor 2 accounts for the energy being split between $+f_0$ and $-f_0$. For $k = 0$ (DC component) the mean value is X_0/N (no factor 2).

This is exactly what the code does in Part 5: it recovers $A_1 \approx 1.005$, $A_2 \approx 0.486$ and phases 0.284 and 1.106 rad (true values: 1, 0.5, 0.3, 1.1), with added noise of amplitude 0.3.

Figure: normal modes of two coupled oscillators



Noisy signal in time (top) and its spectrum (bottom): sharp peaks at 3 Hz and 7 Hz.

20. Cost and the FFT

Evaluating $X_k = \sum_m x_m W_N^{km}$ for every k costs N^2 complex multiplications: $O(N^2)$. **Cooley–Tukey idea** (N even): split m into even ($m = 2r$) and odd ($m = 2r+1$):

$$X_k = \underbrace{\sum_{r=0}^{N/2-1} x_{2r}\, W_{N/2}^{kr}}_{E_k} + W_N^k \underbrace{\sum_{r=0}^{N/2-1} x_{2r+1}\, W_{N/2}^{kr}}_{O_k}, \qquad \text{since } W_N^{2kr} = W_{N/2}^{kr}.$$

E_k and O_k are size-$N/2$ DFTs, periodic in k with period $N/2$. Also $W_N^{N/2} = -1$, so

$$X_k = E_k + W_N^k O_k, \qquad X_{k+N/2} = E_k - W_N^k O_k \qquad (k = 0, \ldots, N/2-1).$$

Result

Recurrence $T(N) = 2T(N/2) + O(N) \Rightarrow T(N) = O(N \log_2 N)$. For $N = 4096$: $\sim 5 \times 10^4$ operations vs $\sim 1.7 \times 10^7$. In the code (Part 4): **0.59 s** (hand-written DFT) vs **0.14 ms** (FFT), numerical difference $\sim 10^{-10}$ (round-off).

The simplest version requires $N = 2^p$; libraries handle any N (slower if N has large prime factors).

21. Spectral leakage and windows

The DFT implicitly assumes the sampled signal **repeats periodically** with period $N\Delta t$.

- If the window contains a *whole* number of periods, the repetition is continuous and the spectrum is a clean delta.
- If not (f_0 is not a multiple of f_s/N), the repetition has an **artificial jump** at the edges, and energy "leaks" into many neighboring frequencies (like Gibbs: a jump $\Rightarrow$ coefficients decaying as $1/k$).

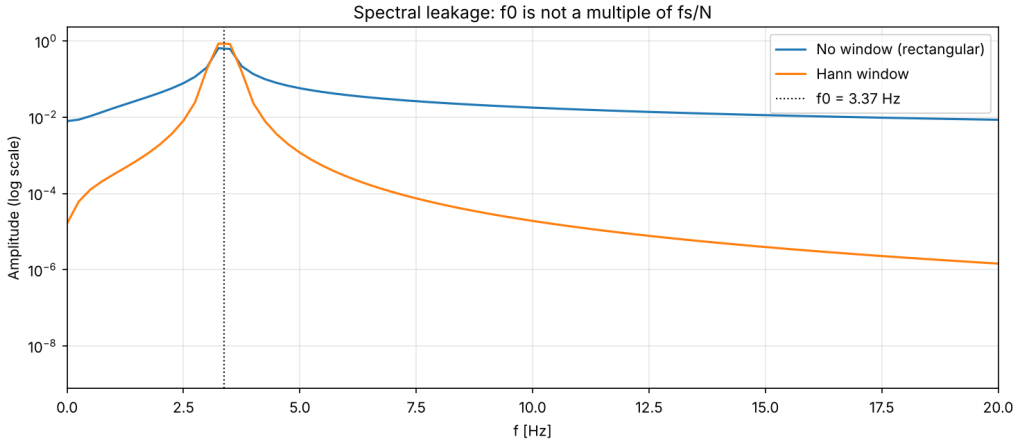
Mathematically: observing N samples equals multiplying by a rectangular window, and a product in time is a **convolution** in frequency with the kernel $\sin(N\pi\Delta f)/\sin(\pi\Delta f)$, whose side lobes decay slowly.

Hann window: $w_m = \frac{1}{2}\left[1 - \cos(2\pi m/N)\right]$

It takes the signal smoothly to zero at both ends, removing the jump. Cost: the main peak widens ($\approx 2\times$). Normalize by $\sum w_m$ instead of N. In the code (Part 6): leakage far from the peak drops from 1.7×10^{-2} to 1.7×10^{-5} (a factor of a thousand).

Rule of thumb: if possible, measure a whole number of periods; if not, apply a window.

Figure: leakage with and without a window



$f_0 = 3.37$ Hz is not a multiple of $\Delta f = 0.25$ Hz. The Hann window drastically reduces the side lobes.

22. Zero padding: finer grid, same resolution

Observing a signal for a time T gives a *continuous* spectrum (the DTFT) whose features have width $\sim 1/T$. The DFT only **samples** it at $f_k = kf_s/N$.

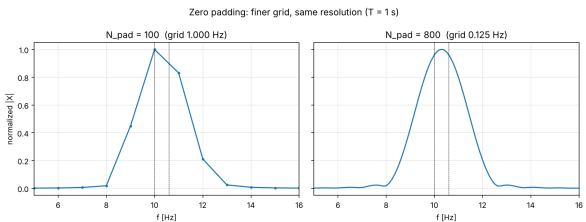
- Appending zeros ($N \to N_{\mathrm{pad}}$) samples the same spectrum on a finer grid f_s/N_{pad}: **interpolation**, no new information.
- The **true resolution** is set by $T = N/f_s$: two tones closer than $\sim 1/T$ stay merged, whatever the padding.
- Useful for: locating a peak more precisely, smooth plots, and making $N_{\mathrm{pad}} = 2^p$ (fast FFT).

Example in the code (Part 9)

Tones at 10 and 10.6 Hz, $T = 1$ s ($1/T = 1$ Hz). With $N_{\mathrm{pad}} = 100$ or 800 they appear as a single peak: padding only draws it smoothly.

To resolve them

Measure longer. Numerically (Hann window), $T = 4$ s already shows two peaks at 10.0 and 10.6 Hz, while $T = 1$ s cannot.



23. The Python code: structure

File: `fourier_physics.py` (each part is an independent, commented function).

Part	What it does	Theory concept
1	Analytic a_n, b_n vs numerical integration (trapezoid)	Orthogonality, odd symmetry
2	Partial sums $S_N(t)$ and overshoot	Convergence, Gibbs
3	$\frac{1}{2}\sum b_n^2 \to 1$	Parseval
4	DFT as matrix Fx vs `np.fft.fft`	Definition, $O(N^2)$ vs $O(N \log N)$
5	Two normal modes + noise; read f, A, φ	Frequencies, physical amplitude
6	Non-integer f_0, no window vs Hann	Spectral leakage
7	Square vs triangle wave decay, Parseval	Smoothness vs decay
8	$N = 4$ by hand vs code	Definition, symmetries
9	Zero padding of two close tones	Resolution vs grid

- **Run:** `python fourier_physics.py` (needs NumPy and Matplotlib). It saves 5 PNG figures.
- **Version note:** in NumPy ≥ 2.0 `np.trapz` was renamed `np.trapezoid`; the code supports both.
- In Part 1, $a_n \sim -10^{-5}$ instead of exactly 0: an endpoint artifact of the trapezoid rule (discretization slightly breaks periodicity at $t = \pm\pi$), not a flaw of the theory.

24. Key code excerpts

DFT as a matrix product (definition, $O(N^2)$):

```
def dft_manual(x):
    N = len(x)
    m = np.arange(N)
    k = m.reshape((N, 1))
    F = np.exp(-2j * np.pi * k * m / N)   # F_km = W_N^(km)
    return F @ x                          # X = F x
```

Spectrum of a real signal with physical amplitudes:

```
X     = np.fft.rfft(x)                    # k = 0 ... N/2
freqs = np.fft.rfftfreq(N, d=1/fs)        # f_k = k * fs / N
amp   = 2.0 * np.abs(X) / N               # A = 2|X_k|/N
amp[0] /= 2.0                             # DC term has no factor 2
phase = np.angle(X)                       # phi = arg X_k
```

Hann window (against leakage):

```
hann = 0.5 * (1 - np.cos(2*np.pi*np.arange(N)/N))
amp_w = 2 * np.abs(np.fft.rfft(x * hann)) / np.sum(hann)
```

25. Numerical results obtained

Part 1 – coefficients (b_n)

n	analytic	numeric
1	1.27323954	1.27323954
3	0.42441318	0.42441318
5	0.25464791	0.25464791
7	0.18189136	0.18189136
9	0.14147106	0.14147106

Part 2 – Gibbs overshoot

N	$\max S_N - 1$
9	0.182328
49	0.179113
199	0.178988
999	0.178980

Theory: $\frac{2}{\pi}\mathrm{Si}(\pi) - 1 = 0.178980$

Part 3 – Parseval (square wave)

N	$\frac{1}{2}\sum b_n^2$
9	0.95960
99	0.99595
999	0.99959
9999	0.99996

Part 5 – normal modes

mode	f [Hz]	A	φ [rad]
1 (true: 1, 0.3)	3.00	1.005	0.284
2 (true: 0.5, 1.1)	7.00	0.486	1.106

Part 7 – triangle Parseval: 3.28986813 vs $\pi^2/3 = 3.28986813$.
Part 8: hand-computed $N = 4$ DFT equals `np.fft.fft` exactly.

26. Analysis checklist for a real signal

1. **Choose f_s** at least 2.5–10 times the highest frequency of interest (Nyquist); low-pass filter if needed.
2. **Choose the duration** T from the required resolution: $\Delta f \approx 1/T$ (and, with a window, about $2\times$ coarser).
3. **Remove the mean (and trend)**, otherwise a huge $k = 0$ peak leaks into nearby bins.
4. **Apply a window** (Hann) unless an integer number of periods is guaranteed.
5. Compute `rfft` and `rfftfreq`; convert with $A = 2|X_k|/N$ (divide by $\sum w_m$ if windowed).
6. **Zero-pad** (optional) for smoother plots or precise peak location; remember it does not improve resolution.
7. **Interpret**: peaks $\leftrightarrow$ normal modes / driving harmonics; check that nothing sits near $f_s/2$ (possible aliasing).

Typical physics uses

Normal-mode frequencies of oscillators, vibration spectra, diffraction and optics (Fourier optics), solving linear PDEs, filtering experimental noise, and spectral analysis of simulation output.

27. Summary and common mistakes

Key ideas

- Fourier series: expansion in an orthogonal basis; coefficients come from projecting.
- Symmetries of f remove coefficients *exactly*.
- Discontinuity $\Rightarrow c_n \sim 1/n$ and Gibbs; smoother $f \Rightarrow$ faster decay.
- DFT = rectangle rule on the complex series: $c_k \approx X_k/N$.
- Resolution $\Delta f = f_s/N = 1/T$; maximum frequency $f_s/2$.
- FFT: same result as the DFT, in $O(N \log N)$.

Common mistakes

- Forgetting the $2/N$ factor for physical amplitude.
- Reading k as a frequency (it is $f_k = k f_s/N$).
- Sampling below $2f_{\max}$ (aliasing).
- Ignoring leakage: window with non-integer periods.
- Thinking zero padding improves resolution.
- Mixing normalization conventions between libraries.
- Reading $k > N/2$ as high frequencies (they are *negative* frequencies).

Suggested references: M. Newman, *Computational Physics*, ch. 7 (Fourier transforms); M. Boas, *Mathematical Methods in the Physical Sciences*, ch. 7; J. W. Cooley and J. W. Tukey (1965), *Math. Comput.* 19, 297.